



Toolset Parameters

AI-A-Carte Cross-Section Tool

Parameters Environments

* Cross-Section Line(s) 1

* Surface Topography DEM 2

* Borehole Points 3

* Borehole Intervals Table (Lithology, Screens, Sampling, etc.) 4

☐ Welllog Data (Unchecked) or Custom Data (Checked) 5

Elevation Units 8

Groundwater Surface(s) 9

Bedrock Surface 10

Other Raster Surfaces 11

Desired Stick Type 12

Polyline 13

Selection Distance 120 Meters 14

Vertical Exaggeration (V.E.) 15

* Profile Grid Options 16

* Output Geodatabase (Not Default GDB)

Geophysical Readings Tool

Parameters Environments

* Cross-Section Line(s) 1

* Elevation Units 8

* Surface DEM 2

* Borehole Points 3

* Well ID Field 17

* Elevation Field 18

* Geophysical Log Data 19

Well ID

Data File

Depth Units

Perform Running Average? ☐

Running Average Value

Selection Distance 120 Meters 13

Vertical Exaggeration (V.E.) 5 14

Horizontal Exaggeration (H.E.) 20

* Output Geodatabase (Not Default GDB) 16

Surficial Markers & Intersections Tool

Parameters Environments

* Cross-Section Line(s) 1

Selection Distance 120 Meters 13

Other Intersecting Features 21

* Surface DEM Profile 2

Elevation Units 8

Feet 8

Vertical Exaggeration (V.E.) 5 14

* Output Geodatabase (Not Default GDB) 16

Surface Profile Tool

Parameters Environments

* Cross-Section Line(s) 1

* Raster DEM Layer 22

Elevation Units 8

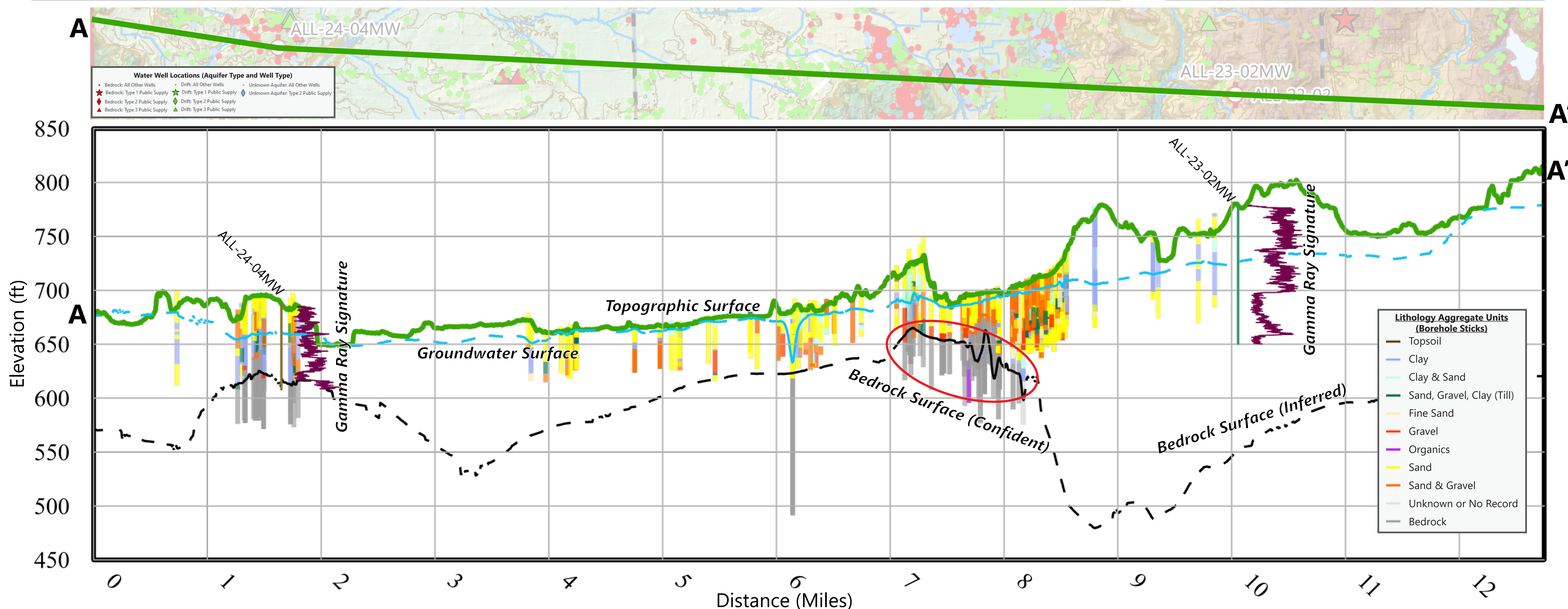
Feet 8

Vertical Exaggeration (V.E.) 5 14

* Output Geodatabase (Not Default GDB) 16

Abstract

Visualizing geologic information in multiple dimensions is critical for making interpretations and informed decisions. Whether informing the public on aquifer storage, generating subsurface models, or presenting spatial information, cross-sections are powerful tools to show what is happening beneath the surface. Advances in computer programming have made subsurface visualizations more accessible, yet many tools remain proprietary and unavailable for general use. The Michigan Geological Survey is developing a suite of cross-section tools that works seamlessly with Michigan's water well database within ArcGIS Pro to visualize water well and borehole data. These tools have also been developed to produce profile views of elevation surface rasters (ranging bedrock tops, formation tops, etc.) as well as single-field specific geophysical logs, such as gamma ray logs, resistivity logs, etc. These tools produce feature class displays in ArcGIS Pro for each cross-section line provided in separate maps. While these tools are being developed for the needs of the Michigan Geological Survey, it is recognized that other organizations have similar needs and could make use of the cross-section tools presented here to work with their own datasets. We have developed and tested our tools to work with a broad range of datasets from multiple states and organizations. We have made the code available on GitHub for other organizations to use and modify for their specific needs. This suite of cross-section tools have allowed team members at the Michigan Geological Survey to more quickly produce cross-sections of borehole data, saving both time and effort for time-sensitive projects, requiring only a well point feature class, lithology or intervals table, cross-section polyline feature class, and a project specific DEM raster.



Tool Descriptions

AI-A-Carte Cross-Section Tool

The following tool produces cross-section view(s) of borehole data, surface topography, groundwater elevation surface, bedrock elevation surface, and gridline generation. If no cross-section has been made previously, the preliminary cross-section tool is meant to produce a first-look at the cross-sections. This combines the Borehole Intervals, Profile, Segmentation of Profiles, and Gridline Creation tools in a single-step process for all known cross-sections.

Geophysical Readings Tool

The following tool takes raw gamma ray log LAS data and transposes the data into a cross-sectional view. Each profile is a separate feature class to display the data. This can be used in conjunction with any of the other MGS Cross-Section tools as a stand-alone script.

Surficial Markers & Intersections Tool

This tool can produce markers on cross-section lines that marks where features intersect the cross-section. This can be used in conjunction with any of the other MGS Cross-Section tools as a stand-alone script.

Surface Profile Tool

The following tool produces a profile based off a raster elevation surface. This can be used in conjunction with any of the other MGS Cross-Section tools as a stand-alone script.

Borehole Interval Sticks Tool

Segment drilling borehole sticks by any user defined interval table (lithology tables, screen tables, sampling intervals, etc.) This can be used in conjunction with any of the other MGS Cross-Section tools as a stand-alone script.

Segmentation of Surface Profiles Tool

This tool produces profiles that are separated by a polygon feature class. The tool requires that a raster elevation surface and a segmenting polygon to run. This can be used in conjunction with any of the other MGS Cross-Section tools as a stand-alone script.

Gridline Creation Tool

The following tool allows for generation of gridded intervals to display the cross-section data. The tool will encompass the data within the boreholes, bedrock elevation surface, and topographic surface rasters. This can be used in conjunction with any of the other MGS Cross-Section tools as a stand-alone script.

Borehole Sticks Tool

The following tool is designed to project subsurface borehole information into a profile view. This can be used in conjunction with any of the other MGS Cross-Section tools as a stand-alone script.

Parameter/Field Hints

- Cross-section line(s):** These are polylines that define where to draw the cross-section. Two fields are required to run, DIRECTIONS and XSEC. DIRECTIONS tells the tool where to read the cross-section line and XSEC determines which line it is reviewing.
- Surface DEM Raster:** The user provided surface topography raster. This can be in any raster format.
- Borehole Points:** The user provided borehole data. If the data is from Welllog, Michigan's water well database, all fields needed are present. Otherwise, refer to **Parameter 5**.
- Intervals Table for Borehole Points:** A user provided table with intervals marked at depth. This can be lithology changes, screened intervals, sampling intervals, etc. If data is from Welllog, all fields needed are present (Well ID, Bottom Depth, Top Depth). Otherwise, refer to **Parameter 5**.
- Boolean check box for data formatting of borehole data. If the box is unchecked, this tells the code that the data is from Welllog. If it is checked, new parameters will become active for the user to define required fields for the tools (See **Parameter 6** and **7**).
- Fields for Borehole Points: The Well ID field and the well depth is required to run the tool. If the user would like to use the elevation data in the borehole database, define the elevation field as well. Otherwise, elevation is taken from DEM. ***NOTE: Depth cannot be null.**
- Fields for Intervals Table: The Well ID field, top depth field, and bottom depth field are all required to run the tool.
- The elevation units for the datasets:** This tells the tool what units are being measured in the depth and elevation values. **THIS MUST BE THE SAME FOR ALL DATASETS.** The default is Feet.
- Groundwater surface(s) (*Only for AI-A-Carte Cross-Sections Tool*): The user can provide the surface of the groundwater raster. If the raster applies to a date range, define the start and end year. If not provided, all wells in the dataset will be applied for confidence zones. This is not required to run the tool.
- Bedrock surface (*Only for AI-A-Carte Cross-Sections Tool*): The user provided bedrock topography surface. This is not required to run the tool.
- Any other raster surfaces (*Only for AI-A-Carte Cross-Sections Tool*): If the rasters do not fit into the Bedrock or Groundwater surface category, the user can have them projected on the cross-section here. The surface still needs elevation values to run.
- The user's desired borehole stick type:** Polyline is designed for large density of boreholes over a long distance. Polygons are designed for small lines with low density of boreholes along the line. Only applies to boreholes. Default is Polyline.
- The selection distance for the tool:** How far away will the tool look from the line to pull in data? This is on both sides of the line (100 meters one direction and 100 meters in the other direction). Default is 120 Meters.
- Vertical Exaggeration (V.E.):** How much does the user want to stretch the vertical component? Default is 50x.
- Parameters for the grid layout:** Such as in a chart, how do you want the output to look? Elevation interval and distance intervals in whole numbers and defining the linear units to display (Feet, meters, miles, kilometers).
- The output geodatabase for the files:** This CANNOT be the same as the default geodatabase. Any other geodatabase is acceptable.
- Well ID Field (*Only for Geophysical Readings Tool*):** User defined Well ID field in the borehole points. The values in this field will be used in Parameter 19.
- Elevation Field (*Only for Geophysical Readings Tool*):** User defined elevation field in the borehole points.
- Geophysical log data:** Well ID - Which well has the geophysical data? The data file - This needs to be in a Microsoft Excel file with two fields, DEPTH and DATA. The data can be anything, but needs a depth component. Depth Units - What is the depth unit in the Excel table? Rolling Average - Checked means yes, unchecked means no. If yes, how much of a rolling average?
- Horizontal Exaggeration (H.E.) (*Only for Geophysical Readings Tool*):** How much does the user want to stretch the relative readings to show them on the cross-section? The default is 1.
- Other intersecting features (*Only for Surficial Features Intersection Tool*): What features do you want to mark on the cross-section line? If nothing is applied, the tool will just mark where cross-section lines intersected each other.
- Raster DEM Layers (*Only for Profiles Tool*): Which rasters need to be projected on a cross-section view? This is the same as **Parameter 11**, but specifically only for the Profiles tool.
- Rasters and polygons to split rasters (*Only for Segmentation of Profiles Tool*):** Define the raster and the polygon the user wishes to split the polygons by. ***NOTE:** The polygons cannot have any holes or overlapping polygons ANYWHERE in the feature class. This will cause the tool to fail.
- The grid extent maximum and minimum elevation (*Only for Gridline Creation Tool*):** The maximum elevation and minimum elevation to build the gridline. This is not vertically exaggerated.

Acknowledgements

A special thanks to those that have provided feedback and tested the tool during the duration of editing. Thank you to Dr. Nathan Erber and John Esch for testing the tools and providing improvements to the process. Thank you to Thomas Valachovics for providing critical insight on the process and workflow recommendations.

We also would like to acknowledge to Evan Thoms from USGS that provided the starting point on GitHub. Without the work done by Evan, the tool would not look like how it is today.

Toolset Parameters

Borehole Interval Sticks Tool

Parameters Environments

* Cross-Section Line(s) 1

* Surface DEM 2

* Borehole Points 3

* Borehole Intervals Table (Lithology, Screens, Sampling, etc.) 4

☒ Welllog Data (Unchecked) or Custom Data (Checked) 5

Borehole Points: Fields 6

ID Field 6

Depth Field 6

Elevation Field (Optional) 6

Borehole Intervals Table: Fields 7

ID Field 7

Depth Top Field 7

Depth Bottom Field 7

Elevation Units 8

Feet 8

Selection Distance 120 Meters 13

Vertical Exaggeration (V.E.) 5 14

Output Style 12

Polyline 12

* Output Geodatabase (Not Default GDB) 16

Segmentation of Surface Profiles Tool

Parameters Environments

* Cross-Section Line(s) 1

* Rasters & Segmentation Polygons 23

Elevation Units 8

Feet 8

Vertical Exaggeration (V.E.) 5 14

* Output Geodatabase (Not Default GDB) 16

Gridline Creation Tool

Parameters Environments

* Cross-Section Line(s) 1

* Elevation Units 8

* Surficial DEM Raster 2

Grid Extent - Elevation 24

Maximum Elevation 24

Minimum Elevation 24

Vertical Exaggeration (V.E.) 5 14

* Grid Options 15

Elevation Increment 15

Elevation Units 15

Distance Increment Value 15

Distance Units 15

* Output Geodatabase (Not Default GDB) 16

Borehole Sticks Tool

Parameters Environments

* Cross-Section Line(s) 1

* Surficial DEM 2

* Borehole Points 3

☐ Welllog Data (Unchecked) or Custom Data (Checked) 5

Elevation Units 8

Feet 8

Selection Distance 120 Meters 13

Vertical Exaggeration (V.E.) 5 14

Stick Type 12

Polyline 12

* Output Geodatabase (Not Default GDB) 16